Correlation

Pierre Pinson

Imperial College London, Dyson School of Design Engineering
mail: p.pinson@imperial.ac.uk - **webpage:** www.pierrepinson.com

Game Theory and Mechanism Design - Lecture 9

(This lecture heavily relies on Narahari (2014) Game Theory and Mechanism Design – chap. 25)

Learning objectives

At the end of this lecture (and related tutorial(s)), a student should be able to:

- Understand different types of **interactions** between agents
- Define and discuss the specifics of **contracts**
- Describe the concept of **correlated strategies**
- Explain and use **individual rationality**

Outline

1. Bringing interaction into the picture
2. Games with contracts
3. Correlated strategies
4. Individual rationality

1. **Bringing interaction into the picture**

Back to our prisoner dilemma example

In principle, when thinking of the prisoner dilemma, we focus on the rational and selfish setup...

		player 2	
		NC	C
player 1	NC	$-4, -4$	$\overline{-6, 0}$
	C	$\underline{0, -6}$	$\overline{\underline{-5, -5}}$

- We can easily spot (C,C) as a **strictly dominant equilibrium**... (and hence, a Nash equilibrium)
- However, many times, we thought that there should be a way for them to get to (NC,NC) which would yield higher pay-offs (in terms of Pareto dominance)!

If only they could interact with each other...

How could they interact?

We generally see 3 main ways of interaction between players

Communication

This is a *light touch approach*, as they (in)directly inform each other about their potential moves

Contract

This somewhat assumes *interaction prior to, or during* the game, to agree on how to play (and it should be binding)

Repetition

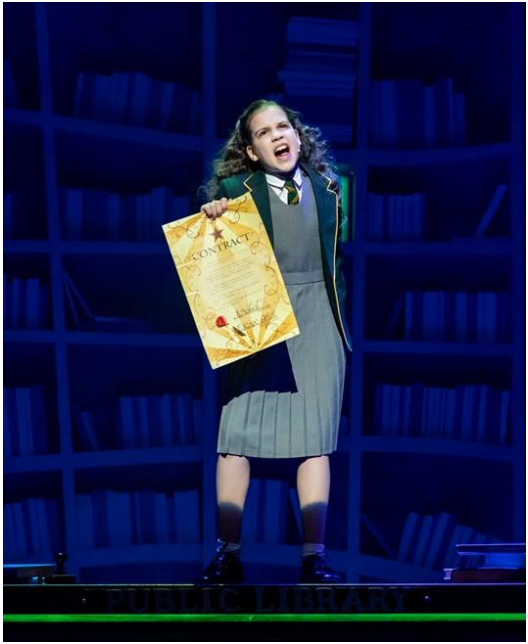
This is *indirect interaction* only, since only observing previous outcomes, and inferring the other's behaviour

Do you see advantages and caveats with these 3 different ways of interaction?

(Note that we will only look at contracts in the following...)

2. **Games with contracts**

A contract is on offer...



In game theory:

- a contract is one type of interaction among players
- it comprises an agreement *prior to, or during* the game
- from a strategic point of view, we consider that players may sign it or not (if they sign it, they respect it!)
- let us say that, if they sign and respect it, they then *apply the contract*

[Matilda - Cory Sincennes (at Cambridge Theatre)]

A proposal contract

		player 2		
		NC	C	AC1
player 1	NC	$-4, -4$	$\overline{-6, 0}$	$\overline{-6, 0}$
	C	$\underline{0, -6}$	$\overline{\underline{-5, -5}}$	$\overline{-5, -5}$
	AC1	$\underline{0, -6}$	$\underline{-5, -5}$	$\overline{\underline{-4, -4}}$

C: Confess
NC: Not Confess
AC1: Apply Contract 1

(If player i alone signs the contract, they know that player j did not sign... hence they would play C)

Let us say that the 2 players have had a discussion in the past, to agree on what they would do if getting caught... (i.e., **not confess!**)

- we refer to is as *Contract 1*
- they now have an *extended strategy set* {C,NC,AC1}
- hence, it is a strategic decision to *apply the contract*, or not...

What is the impact on equilibria?

A proposal contract

		player 2		
		NC	C	AC1
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- they now have an *extended strategy set* {C,NC,AC1}
- hence, it is a strategic decision to *apply the contract*, or not...

What is the impact on equilibria?

- (C,C) is not a strictly dominant equilibrium anymore (it is a weakly dominant equilibrium instead)
- another weakly dominant equilibrium has appeared, i.e., (AC1,AC1)

Could they get even better (expected) pay-offs?

Let me propose an additional contract (*Contract 2*):

- If both players *apply contract 2* (i.e., sign and respect it), a coin is tossed...
  - if "heads", they will implement (NC,C)
  - if "tails", they will implement (C,NC)
- If only player 1 applies contract 2 (hence, knowing that player 2 has not agreed to it), they will play C
- If only player 2 applies contract 2 (hence, knowing that player 1 has not agreed to it), they will play C

Players now have an extended strategy set with 4 strategies: {C,NC,AC1,AC2}

Can you see the way it impacts the game?

Resulting game and equilibria

player 2

player 1	NC	C	AC1	AC2
NC	$-4, -4$	$-6, 0$	$-6, 0$	$-6, 0$
C	$0, -6$	$\mathbf{-5, -5}$	$-5, -5$	$-5, -5$
AC1	$0, -6$	$-5, -5$	$\mathbf{-4, -4}$	$-5, -5$
AC2	$0, -6$	$-5, -5$	$-5, -5$	$\mathbf{-3, -3}$

C: Confess

NC: Not Confess

AC1: Apply Contract 1

AC2: Apply Contract 2

The resulting pure strategy Nash equilibria are:

- (C,C) - with pay-offs (-5,-5)
- (AC1,AC1) - with pay-offs (-4,-4)
- (AC2,AC2) - with pay-offs (-3,-3)

We also have a mixed-strategy Nash equilibrium $\left(\left(0, 0, \frac{2}{3}, \frac{1}{3}\right), \left(0, 0, \frac{2}{3}, \frac{1}{3}\right)\right)$

3. **Correlated strategies**

Coordination of individual strategies

Until now, we have looked at players making decisions individually and independently... trying to see what the outcome may be.

However, based on external factors, past experience, etc., there may be reasons to think their choices for strategies are interconnected.

Definition (Correlated strategies)

Considering a normal form game $\Gamma = \langle N, (S_i), (u_i) \rangle$, and a subset of players $\mathcal{C} \subseteq N$ (a *coalition*). A *correlated strategy* $\tau_{\mathcal{C}}$ is a probability distribution over the set of potential combinations of their individual strategies. We write

- $S_{\mathcal{C}} = \prod_{i \in \mathcal{C}} S_i$ the set of potential combinations of pure strategies for C,
- $\Delta S_{\mathcal{C}}$ the set of probability distributions over $S_{\mathcal{C}}$.

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Example:

- consider 3 players $N = \{1, 2, 3\}$, but only the last 2 as a coalition $\mathcal{C} = \{2, 3\}$
- they have strategies $\{a_1, b_1, c_1\}$ (player 1), $\{a_2, b_2, c_2\}$ (player 2) and $\{a_3, b_3\}$ (player 3)
- the set of combinations of pure strategies (for $\mathcal{C}$) is

$$S_{\mathcal{C}} = \{(a_2, a_3), (a_2, b_3), (b_2, a_3), (b_2, b_3), (c_2, a_3), (c_2, b_3)\}$$

- an example correlated strategy is

$$\tau_{\mathcal{C}} = \left(\frac{1}{12}, \frac{1}{12}, \frac{1}{12}, \frac{1}{3}, \frac{1}{4}, \frac{1}{6}\right)$$

Expected pay-offs

To make things simpler, we consider that $\mathcal{C} = N$ (referred to as *grand coalition*)

Definition (Expected pay-offs from correlated strategies)

For a normal form game $\Gamma = \langle N, (S_i), (u_i) \rangle$, a grand coalition $\mathcal{C} = N$ and a correlated strategy $\tau_{\mathcal{C}}$, the *expected pay-off* of player i is

$$u_i(\tau_{\mathcal{C}}) = \sum_{s \in S} \tau_{\mathcal{C}}(s)\, u_i(s)$$

where s is a given strategy profile within $S = \prod_i S_i$.

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Example:

		player 2	
		NC	C
player 1	NC	$-4, -4$	$-6, 0$
	C	$0, -6$	$-5, -5$

$S_{\mathcal{C}} = \{(NC, NC), (NC, C), (C, NC), (C, C)\}$

Consider a correlated strategy $\tau_{\mathcal{C}} = \left(\frac{1}{8}, \frac{1}{4}, \frac{1}{4}, \frac{3}{8}\right)$

The expected pay-off of player 1 is

$$u_1(\tau_{\mathcal{C}}) = \frac{1}{8} \times (-4) + \frac{1}{4}(0 - 6) + \frac{3}{8} \times (-5) = -\frac{31}{8}$$

For player 2, we also have

$$u_2(\tau_{\mathcal{C}}) = -\frac{31}{8}$$

(These expected pay-offs are better than any of the pay-offs from implementation of individual strategies!)

Definition (A contract)

For a normal form game $\Gamma = \langle N, (S_i), (u_i) \rangle$, coalitions $\mathcal{C} \subseteq N$ and correlated strategies $\tau_{\mathcal{C}}$, a *contract* $\tau = (\tau_{\mathcal{C}})_{\mathcal{C} \subseteq N}$ is a vector of correlated strategies for all potential coalitions.

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Examples:

Prisoner dilemma with Contract 1

- we have 3 potential coalitions $\{1\}$, $\{2\}$ and $\{1, 2\}$
- For $\mathcal{C} = \{1\}$, we have $\tau_1 = (0, 1)$ (i.e., probability 0 of playing C, since strictly dominated)
- For $\mathcal{C} = \{2\}$, we have $\tau_2 = (0, 1)$ (i.e., probability 0 of playing C, since strictly dominated)
- For $\mathcal{C} = \{1, 2\}$, we have $\tau_{\{1,2\}} = (1, 0, 0, 0)$ (i.e., probability 1 of playing (NC,NC))

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- For $\mathcal{C} = \{1, 2\}$, we have $\tau_{\{1,2\}} = (1, 0, 0, 0)$ (i.e., probability 1 of playing (NC,NC))

Prisoner dilemma with Contract 2

- we have 3 potential coalitions $\{1\}$, $\{2\}$ and $\{1, 2\}$
- For $\mathcal{C} = \{1\}$, we have $\tau_1 = (0, 1)$ (i.e., probability 0 of playing NC, since strictly dominated)
- For $\mathcal{C} = \{2\}$, we have $\tau_2 = (0, 1)$ (i.e., probability 0 of playing NC, since strictly dominated)
- For $\mathcal{C} = \{1, 2\}$, we have $\tau_{\{1,2\}} = \left(0, \frac{1}{2}, \frac{1}{2}, 0\right)$ (i.e., probability $\frac{1}{2}$ of playing (C,NC) and (NC,C))

And, the pay-offs were given in the matrix representations shown before...

4. **Individual rationality**

Why would you ever consider signing a contract??

(remember that you are *individualistic*, *selfish* and *rational*...)

Security value

Players would use as a reference the worst that could happen to them, if not having any interaction

Definition (Security value)

In a normal form game $\Gamma = \langle N, (S_i), (u_i) \rangle$, the *security value* v_i of player i is the minimum guaranteed pay-off if freely choosing any strategy $s_i \in S_i$ as a response to the strategy of others ($s_{-i} \in S_{-i}$), i.e.

$$v_i = max_{s_i \in S_i} \, min_{s_{-i} \in S_{-i}} \, u_i(s_i, s_{-i}) \, .$$

Owing to this formulation, it is also referred to as *maxmin* value. The strategy s_i that yields v_i is the *security strategy*.

(Interpret it as "*even if the other players were trying to hurt me, this is the minimum pay-off I could reach*")

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In this prisoner dilemma game, what are the security values v_1 and v_2 of the 2 players?

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	C	$0, -6$	$-5, -5$

In this prisoner dilemma game, what are the security values v_1 and v_2 of the 2 players?

- $v_1 = -5$
- $v_2 = -5$

For both players, C is the security strategy.

Individual rationality

A player i will only apply a contract implying a correlated strategy if $u_i(\tau_{\mathcal{C}}) \geq v_i$... This is referred to as **individual rationality**!

Definition (Individually rational correlated strategy)

A correlated strategy $\tau_{\mathcal{C}} \in \Delta S_{\mathcal{C}}$ is said to be *individually rational* for all players $i \in \mathcal{C}$ if the expected pay-offs from the correlated strategy are greater than their security values, i.e.,

$$u_i(\tau_{\mathcal{C}}) \geq v_i, \quad \forall i \in \mathcal{C}\,.$$

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For instance, for the 2 versions of the prisoner dilemma with contracts, the related correlated strategies are individually rational.

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For instance, for the 2 versions of the prisoner dilemma with contracts, the related correlated strategies are individually rational.

Proposition (Nash equilibrium in contract signing game)

If seeing the possibility of players applying (i.e., signing and respecting) contracts as a game, whenever a correlated strategy is individually rational, this defines a contract for which the actions of all players applying it in a Nash equilibrium.

For 2 versions of the prisoner dilemma with contracts, (AC1,AC1) and (AC2, AC2) are Nash equilibria.

Wrapping up

- Players can be rational and selfish, but still **interact**
- Interaction is a way to **influence outcomes**, and contracts enforce interaction
- **Contracts** only make sense if related to individually rational correlated strategies
- If not possible to use contracts, **Communication** (and mediation) can be used instead (read Narahari (2014) - chap. 25)
- This is our first step towards **cooperative games!**

Further readings (/lecture notes for this topic):

Narahari Y (2014) *Game Theory and Mechanism Design* - chap. 25 (available on Blackboard Learn)

Thanks for your attention!